Hamza Houir Alami

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Engineering student at Arts et Métiers ParisTech (ENSAM),a leading French engineering Grande École. Founder of **ThePnLab**, a live algorithmic trading simulation platform with 37 users. Background in mathematical optimization, consulting, and entrepreneurial product development — targeting graduate studies in **Operations Research** to apply optimization methods at the intersection of finance, supply chain, and technology.

Education

- **Arts et Métiers ParisTech — ENSAM** Lille, France
  MEng in General Engineering (Diplôme d'Ingénieur) *Sep. 2025 – Jun. 2028 (Expected)*
  - **Top 10** French Graduate Engineering School (Grandes Écoles)
  - **Relevant Coursework:** Supply Chain & Logistics, Industrial Optimization, Production Systems, Numerical Simulation (FEM, Python), Fluid Mechanics, Industrial Organization
- **Lycée Faidherbe** Lille, France
  CPGE PSI — Intensive preparation in Physics, Mathematics & Engineering Science* *Sep. 2023 – Jun. 2025*
  - **Top 10** engineering preparatory school in France
  - **Relevant Coursework:** Advanced Mathematics (Analysis, Algebra, Linear Algebra), Physics (Mechanics, Thermodynamics), Engineering Science, Algorithms & Python

Work and Leadership Experience

- **Business Manager — AMJE Lille (Junior Enterprise, Arts et Métiers)** Dec. 2025 – Present
  Student-led engineering consultancy *Lille, France*
  Develop leadership, project management and client relations while supervising OR consulting missions.
  - Led routing optimization mission for a regional distributor serving **650 retail delivery points** (Hérault dept.): modeled as a **VRP**, solved with K-Means++ zone clustering + 2-opt local search, OSRM API for real-road distances — deployed on GitHub Pages
  - Managed full commercial funnel: 286 prospects contacted → 7 studies signed (2.4% conversion); identified bottleneck at qualification stage (14.7% stage yield); designed redesigned outreach strategy targeting +33% revenue growth
- **Industrial Engineering Intern — Safran, Tooling Methods Dept. (Incoming)** Jun. 2026 – Jul. 2026
  Aerospace manufacturing *France*
  Contribute to process engineering and tooling methods in an industrial aerospace environment.
  - Incoming internship: process optimization, tooling design and methods engineering for aircraft component manufacturing
- **Private Tutor — Mathematics & Physics** 2024 – Present
  Self-Employed *Lille, France*
  Competitive exam preparation for baccalauréat and CPGE-level students.
  - Adapted teaching methods to individual learning profiles; focused on analytical reasoning and exam strategy

Selected Research & Development Projects

- **ThePnLab — Algorithmic Paper Trading Platform** 2025 – Present
  thepnlab.com | github.com/Hamza-67/ThePnLab *Individual Project*
  Build a full-stack platform for algorithmic trading simulation; study decision-making divergence between bot and human behavior.
  - Daily decision cycle: composite score on **9 technical signals** selects positions across 36+ assets; **Kelly Criterion** sizes them; **VIX regime controller** scales exposure; LLM (Gemini 2.0 Flash) generates structured trade rationale
  - **Stack:** Python, FastAPI, Streamlit, PostgreSQL — **37 registered users**, live at thepnlab.com; 40-day baseline: bot +0.00% vs. human −5.02% (behavioral divergence confirmed)
- **Flywheel Shape Optimization (TIPE — ENSAM Lille)** 2025
  Mathematical Optimization | Python (SciPy, NumPy) *Individual Research · Score 8/10*
  Formulate and solve a constrained shape optimization problem; validate computationally and experimentally.
  - Maximized moment of inertia $I = \mu \int r^3 \, d\theta$ under fixed-perimeter constraint; KKT conditions and the isoperimetric inequality analytically confirm the **circle as unique global maximizer**
  - Implemented and benchmarked **4 parameterizations** (Fourier, polynomial, polar, 3D surface) using **L-BFGS-B** and **Trust-Region** solvers (`scipy.optimize`); identified Runge instability above polynomial degree 20
  - Experimental validation on a 34g miniature vehicle: **1.8%–5.7% braking distance error** vs. theory ($n = 5$ runs, $p < 0.0001$)

Technical & Language Skills

Programming: Python (NumPy, SciPy, Pandas, Keras), FastAPI, Streamlit, PostgreSQL, Git

Software: CATIA / 3DEXPERIENCE, Star-CCM+, Abaqus, LaTeX

French: Native

Arabic: Native

English: Fluent

Spanish: Intermediate (B1)

Hobbies & Interests

Football: Competed at international youth level — **Ibercup Cascais** tournament (Portugal), **40+ nations**, 100+ teams; coached an amateur team during injury; developing tactical and data-driven analysis of the game.

Geopolitics: Regular reading and self-directed research on international affairs, regional dynamics (Middle East, North Africa, European politics) and global power shifts — building an analytical lens beyond the engineering curriculum.